

Volkan Sarp

☎ +90-536-541-8989 ✉ theneutron@gmail.com 🌐 Google Scholar 🌐 ResearchGate
Türkiye · Open to Relocation · English: Full Professional Proficiency · No Travel Restrictions

Professional Profile

Researcher in solar–terrestrial physics with a background in the observation, simulation, and data analysis of the ionosphere–thermosphere system’s response to solar forcing. Current research focuses on developing a Complexity Analytics framework by integrating and adapting modern analytical methodologies to complex geophysical datasets.

Core Competencies

Solar–Terrestrial Physics: Solar cycle variability, solar active regions, solar flares, space weather effects on the near-Earth environment, ionosphere–thermosphere coupling, travelling atmospheric disturbances

Observational Analysis & Physics-Based Modeling: GNSS TEC data analysis, physics-based numerical simulations, spatiotemporal disturbance characterization

Complexity Analytics Nonlinear time-series analysis, Multi-scale analysis, Wavelet analysis, Multifractal analysis, Data-driven dynamical systems

Education

Ph.D. in Space Sciences and Technologies

2018–2024

Akdeniz University

Antalya, Türkiye

- Thesis: *A Study on the Response of the Thermosphere–Ionosphere System to Solar Flare*
- Combined GNSS-derived ionospheric observations, general circulation model simulations (GITM), and model–observation validation to investigate solar flare impacts on the ionosphere–thermosphere system.
- Identified longitudinal asymmetries in ionospheric disturbances and proposed a geomagnetic field-dependent mechanism (*Space Weather*, 2024).

M.Sc. in Space Sciences and Technologies

2015–2018

Akdeniz University

Antalya, Türkiye

- Thesis: *Analysis of Chaotic Features and Modelling of Sunspot Cycle*
- Developed a nonlinear, data-driven prediction of Solar Cycle 25 using state-space reconstruction (*MNRAS*, 2018).

B.Sc. in Industrial Engineering

2007 – 2013

TOBB University of Economics and Technology

Ankara, Türkiye

- Optimization, time-series analysis, systems modeling, CAD (AutoCAD); cooperative education program with in-field training at multiple industrial companies.

Selected Publications

Sarp, V., Yiğit, E., & Kilcik, A. (2024). Response of the thermosphere-ionosphere system to an X-class solar flare: 30 March 2022 case study. *Space Weather*, 22. doi:10.1029/2024SW003938

Kilcik, A., **Sarp, V.**, Yurchyshyn, V., & Rozelot, J.P. (2020). Physical characteristics of umbral dots derived from high-resolution observations. *Solar Physics*, 295(58). doi:10.1007/s11207-020-01618-y

Sarp, V., Kilcik, A., Yurchyshyn, V., Rozelot, J.P., & Ozguc, A. (2018). Prediction of solar cycle 25: A non-linear approach. *Monthly Notices of the Royal Astronomical Society*, 481(3), 2981–2985. doi:10.1093/mnras/sty2470

10 peer-reviewed journal articles · 4 international conference proceedings — full list available on Google Scholar

Awards & Fellowships

TÜBİTAK Research Fellowship — George Mason University, USA

2025

TÜBİTAK Graduate Scholarship — Akdeniz University, Türkiye

2019 – 2023

SCOSTEP Visiting Scholar Award — NASA Goddard Space Flight Center, USA (awarded; deferred)

2020

Research Experience

Post-Doctoral Research Fellow (TÜBİTAK)

Jan 2025 – Sep 2025

George Mason University

Fairfax, VA, USA

- Developed a multi-scale detrending pipeline for GNSS-TEC observations to characterize scale-dependent ionospheric responses to the April 2024 total solar eclipse.
- Designed an event-centred moving spatial reference frame aligned with the eclipse shadow trajectory at ionospheric pierce-point altitude, enabling coherent isolation and analysis of eclipse-induced TEC perturbations.

Research Scholar (TÜBİTAK): Fine-Scale Dynamics in Sunspot Umbra

2017 – 2019

Akdeniz University

Antalya, Türkiye

- Adapted an IDL-based Lagrangian particle-tracking algorithm for automated detection and lifetime tracking of umbral dots, enabling one of the largest statistical analyses of GST/BBSO observations reported at the time (*Solar Physics*, 2018).
- Showed that mean umbral-dot velocities are nearly twice previously reported values by distinguishing trajectory path length from net displacement, revealing nonlinear wandering motion rather than radial inward propagation.

Scientific Research Project (BAP): Nonlinear Analysis and Prediction of Solar Activity 2017 – 2018

Akdeniz University

Antalya, Türkiye

- Developed nonlinear time-series analysis and state-space reconstruction methods to investigate the dynamics and predictability of solar cycle variability.
- Produced one of the earliest successful data-driven predictions of both the amplitude and timing of Solar Cycle 25 peak using nonlinear state-space reconstruction techniques, later confirmed by observations (*MNRAS*, 2018).